CHAPTER-1

The magnitude of maximum displacement to one side from its equilibrium position is amplitude.

The time taken for one oscillation is called period.

The number of oscillations in one second is called frequency.

When the length of the pendulum increases, frequency decreases.

As the period increases, frequency decreases.

Factors that influence the natural frequency of an object :

Length of the object/Size of the object/ Elasticity/Nature of the material etc

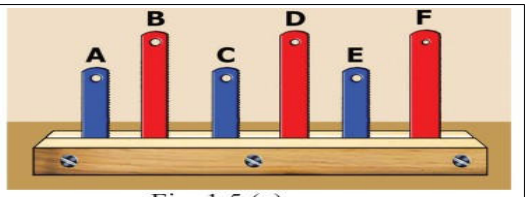


Fig. 1.5 (a)

- Excite the hacksaw blade A by tapping with your finger. What do you observe?
  (all blades vibrate / only A vibrates)
- Are all the blades vibrating with the same amplitude?
- Which of them vibrates with maximum amplitude?
- After all the blades have stopped vibrating, excite B and record the observation in the science diary.

Only A vibrates/No, all the blades are not vibrating with the same amplitude/Blade A vibrates with maximum amplitude.

Applications of forced vibration and resonance

MRI scanning/Radio tuning/In musical instruments like guitar, violin, veena, harmonium, mridangam etc.

The frequency of a simple pendulum is 1 Hz. What is its period?

T=1/f

1/1=1s

If a pendulum takes 0.5 s to complete one oscillation, what is its frequency?

f=1/T

1/0.5=2Hz

Radio waves, microwaves, infrared rays, visible light, ultraviolet rays, X-rays and gamma rays are electromagnetic waves. They do not require a medium for transmission.

Longitudinal waves are those in which the particles in the medium vibrate parallel to the direction of propagation of the wave.

When the particles of a medium vibrate perpendicular to the direction of propagation of the wave they are called transverse waves.

Electromagnetic waves are transverse waves

The main characteristics of waves are :Amplitude/Frequency/Period/Wavelength

Speed of wave.

When the speed is constant, frequency of the wave is inversely proportional to the wavelength.

- What is the wavelength (λ)?
- If the wave takes 1 s to reach A from O, what is the frequency (f)?
- Isn't the speed of a wave the distance travelled by it in one second? What is the speed of the wave (v)?

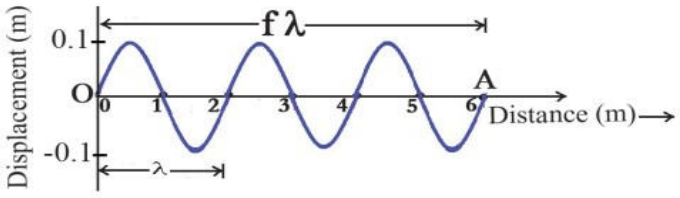


Fig.1.15

The wavelength is 2m/ 6/2=3 cycles/ speed=distance/time=6/1=6m/s

Reflection of sound is utilised in :

Soundboards /Curved ceilings in halls

For the echo to be heard, the reflecting surface must be at least 17.5 m.

frequency below 20 Hz is infrasonic. Sound with frequency more than 20000 Hz is ultrasonic.

Uses of Ultrasonic Waves

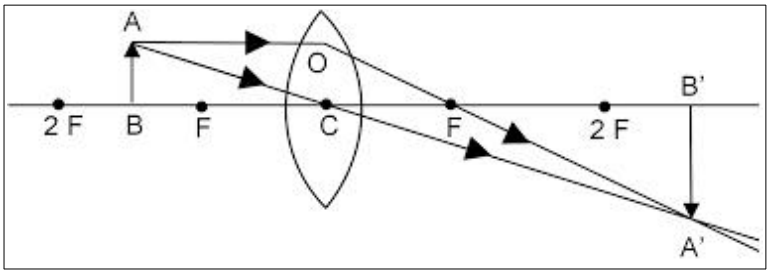
To crush small stones in the kidneys./In physiotherapy.

CHAPTER-2

The focal length (f) is the distance from the optic centre of the lens to the principal focus.

Write down examples for real images.

Image that is captured on a camera/ Image that is formed on a cinema screen



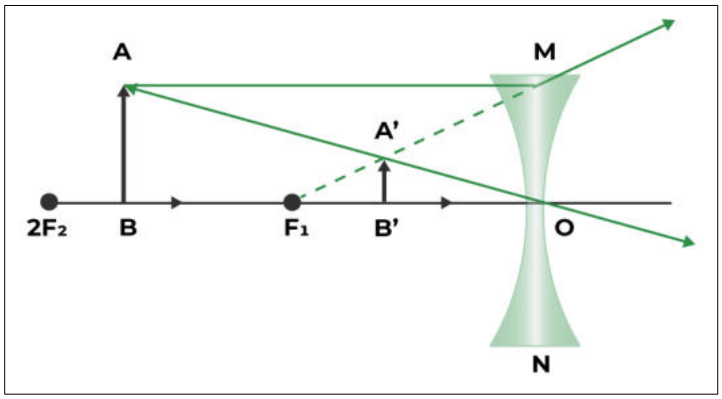
object placed between F and 2F

Position: Beyond 2F

Nature: Real and Inverted

Size: Magnified

Images that cannot be captured on a screen, but can only be seen are virtual images.



Nature: Virtual and Erect
Size: Diminished
Position: Between the optical center and the focal point

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$f = \frac{uv}{u - v}$$

Power is the reciprocal of focal length.
The image we see through the eyepiece is virtual in telescope

CHAPTER-3

What are the factors on which the deviation of a ray of light depend?
Refractive index of the medium/Wavelength of the colour of light.
Dispersion of light is the phenomenon of splitting up of a composite light into its component colours.
The orderly distribution of electromagnetic radiations is known as the electromagnetic spectrum.

Overlapping colours	Resulting colours on overlapping
Red + Green	Yellow
Red + Blue	Magenta
Blue + Green	Cyan
Red + Green + Blue	White

What are the primary colours in yellow light?red and green
• Which primary colour is not present in yellow light?blue.
When a burning incense stick is whirled very fast, a ring of fire can be seen. Why?
Persistence of Vision.
What colours do a green leaf reflect?Green leaves primarily reflect green light
And what about a red flower?Red flowers reflect red light

Scattering is the irregular and partial directional deviation of light when it encounters particles in a medium.

The distance that the sun's rays travel through the atmosphere to reach the Earth during sunrise and sunset compared to other times is more.

Which colours undergo the least scattering at these times?red.

the minimum distance for clear vision is 25 cm.--near point.

Short sightedness can be rectified using a concave lens with suitable power.

Long sightedness can be rectified using a convex lens with suitable power

<u>light pollution</u>

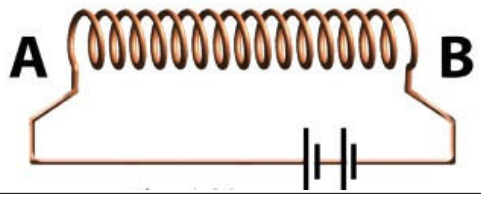
Causes difficulty during night drive./Makes astronomical observations difficult by obstructing the night sky./The light from multi-storeyed flats misleads migratory birds.

CHAPTER-4

If the current in the coil is clockwise, the direction of the flux lines will be inward into the coil. If the current is anticlockwise, the direction of the flux lines will be outward.

When the number of turns of the conductor increased, both the magnetic strength and the magnetic flux increased, but the flux produced by a single turn of the conductor did not increase.

Electromagnets are devices that create magnetic field using electricity.



(a) What is the magnetic polarity of end A?

(b) What is the magnetic polarity of end B?

A-North pole

B-South pole

If the direction of the current and the magnetic field are reversed together, the conductor will move in the same direction as before.

What are the factors that influence the direction of the force experienced by the conductor?

The direction of the current The direction of the magnetic field/The orientation

A current carrying conductor which is free to move, placed in a magnetic field, exhibits a tendency to deflect. This is motor principle.